

EDUCATION

Master of Science in Applied Computing (MScAC) | University of Toronto

September 2026 – December 2027

Department of Computer Science, Faculty of Medicine • Focus in Artificial Intelligence

Courses: Neural Networks and Deep Learning, Computational Imaging and 3D Sensing, Automated Reasoning with Machine Learning

Honours Bachelor of Science (HBSc) | University of Toronto

September 2022 – June 2026

Department of Computer Science • Major in Computer Science, specialist in Bioinformatics & Computational Biology

GPA: 3.9/4.0 • Dean’s List Scholar (2022 – 2026) • B. Arthur Bensley Scholarship in Life Sciences (2025)

Courses: Machine Learning, Artificial Intelligence, Systems Biology, Data Structures and Algorithms, Probability and Statistics

RESEARCH EXPERIENCE

Machine Learning Researcher | Princess Margaret Cancer Centre, UHN

September 2025 – Present

Computational Cancer Genomics Lab, Dr. Sushant Kumar: Automated Segmentation of IHC and H&E Staining

- Designed a multi-modal cascaded diffusion model integrating four structured-data foundation models to synthesize high-resolution images from tabular multi-omics inputs, achieving an 83% match in morphological diversity across image classes using PyTorch
- Led development of a 24.4M-parameter U-Net segmentation pipeline and image-analysis toolkit extensions, trained on 10M image patches for automated structure segmentation and multi-class object quantification, achieving an expert-validated 92% Dice score
- Built and deployed a Python package for large-scale scientific data engineering, delivering ML-ready preprocessing at HPC scale 26% faster than current industry standard benchmarked research toolkits, with automated reproducibility audits and configurable logging

Computer Vision Research Intern | Lunenfeld-Tanenbaum Research Institute, Mount Sinai

May 2025 – February 2026

Campbell Lab, Dr. Kieran Campbell: Computer Vision Methods for Highly Multiplexed Biological Images

- Built an end-to-end image analysis toolkit (ResNet-34 encoder convolutional autoencoder) optimized for highly multiplexed, high-dimensional imaging data, reducing runtime by 30% over existing non-optimized toolkits using Python and PyTorch; cited at *Nature*
- Designed a novel stratified, morphology-specific information saturation sampling protocol for benchmarking imaging modalities against ground truth, capturing the spatial heterogeneity in complex tissue-like environments using PyTorch Lightning; cited at *Nature*

Data Science Research Intern | Massachusetts General Hospital, Harvard Medical School

September 2024 – July 2025

Sadreyev Lab, Dr. Ruslan I Sadreyev: Molecular Mechanisms of Acute Myeloid Leukemia

- Engineered multi-modal sequencing data workflows integrating the largest cross-cohort datasets assembled for the task, applying unsupervised clustering (fuzzy K-means, K=3 and K=5) for integrative differential subtype analysis using R

PROFESSIONAL EXPERIENCE

Lead Systems Software Engineer | Victoria University, University of Toronto

May 2024 – Present

- Led end-to-end design, development, and deployment of a university-scale intranet platform, encompassing architectural redesign, full database restructure and migration, data retention protocol design, and Microsoft 365 migration using PostgreSQL and MongoDB
- Redesigned backend systems architecture and normalized database schema, reducing storage redundancy by 17% using PostgreSQL

SELECTED RESEARCH OUTPUTS

- αSMA-Fibroscore** (Python) – Tuned U-Net for image segmentation and multi-class object quantification/classification pipeline
- WSI-IMC-Workflows** (Python) – Toolkit for tissue/image mask generation, patch extraction, and preprocessing of multiplexed images
- MethylTrain** (Python) – Python package for large-scale structured data engineering for research-grade machine learning workflows
- BioTokenizerR** (R) – R package for domain-aware BPE tokenization of sequencing data optimized for large-scale biomedical datasets

PUBLICATIONS

- Nowlan F, et al. (incl. Li SM), “Integrated spatial proteomics of human PDAC uncover an expanded tumour-immune-stroma spectrum with genomic associations,” *bioRxiv*, 2025 – under second-round peer-review at *Nature*
- Illamperuma AT, Li SM, Ghumman J, Gandhi B, "Validation and algorithmic fairness in AI-enabled ECG models for myocardial ischemia: A systematic review of North American evidence," *Journal of Biomedical Informatics*, 2026 – under first-round peer-review
- Shakfa N, et al. (incl. Li SM), “Single cell imaging uncovers a coordinated tumor-immune-stroma spectrum with genomic associations in PDAC,” *Cancer Research: AACR Annual Meeting*, 2026

TECHNICAL SKILLS

ML/DL: Fluent in PyTorch Lightning, TensorFlow, scikit-learn • Experienced with Machine Learning, Computer Vision, Deep Learning

Languages: Fluent in Python, R, C/C++, Java, and PostgreSQL • Experienced in MySQL, TypeScript, JavaScript, Bash, Assembly